

Contact

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vinaykumarkv.github.io/ (Portfolio)

www.mygreatlearning.com/eportfolio/vinay-kumar-k-v (Portfolio)

Top Skills

Content Creation

Community Building

Curriculum Development

Certifications

SQL Associate

Data Engineer

Business Analysis Foundations

Project Management Essentials
Certified

Professional Scrum Master™ I (PSM I)

Publications

VATSA: Video, Audio, Text, Sensory, Action - A Unified Five-Modality Architecture for Human-Level Perception and Action

Vinay Kumar .K .V

AI Engineer · LLM Systems · Multimodal AI · GxP Regulated Environments | Building VATSA | DBA in AI & ML
London Area, United Kingdom

Summary

I build AI systems that work in the real world - including the most demanding environment you can work in: GxP-regulated pharmaceutical manufacturing. I have grown from being System Engineer -> Product Engineer (Scrum Master, Technical BA, Validation Lead) -> AI Engineer (Technical BA, Product Lead) delivery high value digital products integrated with AI. At London, I don't just evaluate AI. I deploy it. I've built Python utilities running in live production, designed LLM evaluation frameworks that shaped regulatory document automation strategy across US and UK sites, and authored the full GxP validation documentation that got them signed off. But I'm not just a regulated-AI specialist. I'm an AI engineer at my core. I'm independently building VATSA - a unified five-modality AI architecture (Video, Audio, Text, Sensory, Action) for embodied systems. The visual module is complete: 96.31% on CIFAR-10, 22 FPS real-time detection. The audio module hits 70.25% on ESC-50 via Wav2Vec2 transfer learning. I've proposed SAMOS, a novel safety-weighted output routing mechanism for safe parallel multi-modal generation - because a robot must never harm a human, even accidentally. My doctoral research at Walsh College focuses on AI models for safety-critical autonomous systems. The long game is robotics. What I bring that most AI engineers don't: 8 years of understanding how regulated businesses make decisions, how to frame the right problem before writing code, and how to explain technical trade-offs to a board. I know what it takes to get AI trusted and deployed - not just built. Currently open to AI/ML Engineer, Applied AI, and GenAI roles. Technical stack: PyTorch · LangChain · RAG · Azure OpenAI · Docker · FastAPI · FAISS · Scikit-learn · SHAP

Experience

Tata Consultancy Services

10 years 10 months

AI Engineer - LLM & Regulated AI Systems

May 2023 - Present (3 years 3 months)

79 New Oxford St, London WC1A 1DG

LLM Evaluation & Production Deployment

- Defined extraction-quality metrics and led comparative evaluation of Azure Document Intelligence vs GPT-4o Vision for regulatory document automation across US and UK pharmaceutical manufacturing sites — evaluation directly shaped GSK's LLM adoption strategy
- Built and deployed Python XML digital-signature verification utilities into live GxP production; authored full URS/SRS/UAT validation documentation and managed regulatory sign-off end-to-end
- Independently prototyped Azure Document Intelligence and GenAI integrations beyond formal scope; presented technical capabilities and limitations to business stakeholders and engineering leadership, enabling informed adoption decisions

Intelligent Automation & Analytics

- Built Python/Jira API sprint analytics pipeline with automated reporting and visual dashboards — eliminated all manual reporting effort for a 25+ member cross-functional team
- Designed COA digitisation requirements in collaboration with QA, Operations, IT, and Regulatory stakeholders across multiple global sites

Cross-Functional Technical Leadership

- Facilitated workshops across QA, Operations, IT, and Regulatory teams - translated AI evaluation findings into business recommendations that secured executive buy-in for regulatory document automation initiative
- Acted as primary bridge between business requirements and technical delivery, ensuring AI tooling met both operational needs and GxP compliance standards throughout the product lifecycle

Product Engineer - GxP

September 2019 - May 2023 (3 years 9 months)

Bengaluru

Product Engineering & Delivery

- Led cross-functional product team of 8+ developers on Batch Release Hub
- a GxP-validated digital platform replacing paper-based pharmaceutical manufacturing workflows across 8 global sites (US, UK, Europe)
- Translated complex regulatory requirements into sprint-ready technical specifications with measurable acceptance criteria - reduced rework across 8+ major releases
- Owned GxP validation documentation end-to-end throughout the full product development lifecycle: IQ, OQ, PQ protocols, UAT execution, and regulatory sign-off

Platform Transition & Knowledge Engineering

- Led full transition from development to service support including structured knowledge transfer across teams - received formal customer appreciation from GSK for delivery quality

Dual Technical & Process Ownership

- Managed simultaneous Scrum Master and Technical BA responsibilities: sprint planning, backlog management, release coordination, and daily stakeholder communication across global teams

MBA Market Research and projects - Sabbatical

October 2017 - September 2019 (2 years)

Bangalore Urban

Full-time MBA in Marketing & HR at Ramaiah Institute of Technology, Bangalore. Focused on market research, consumer behaviour, and organisational strategy.

Systems Engineer — Enterprise Infrastructure

October 2015 - October 2017 (2 years 1 month)

1/G1, SIPCOT IT Park Navalur, Siruseri, Tamil Nadu 603103, India

- Managed enterprise backup and restore infrastructure across Linux and Windows environments - maintained 100% SLA compliance

- Built shell scripting automation tools to reduce manual operations and improve incident response time

learnwithvinay

Founder and Educator

May 2026 - Present (3 months)

London Area, United Kingdom

Founded and run LearnWithVinay.org - a free, open technology education initiative making AI, data science, mathematics and robotics accessible to non-technical audiences globally

Produce structured curriculum content in personal time: Python Programming (recording in progress), SQL & Databases, Mathematics for Data Science, Machine Learning & Deep Learning, Generative AI, ROS2 & Robotics, Embedded C++ & PCB Design

All content produced voluntarily, entirely independent of professional employment - no fees, no paywalls, no advertising

Education

Walsh College

Doctor of Business Administration, Artificial Intelligence and Machine Learning · (December 2025 - December 2028)

Texas McCombs School of Business

Post Graduate Program in Artificial Intelligence and Machine Learning , Business Applications · (December 2025 - December 2026)

Ramaiah Institute Of Technology

Master of Business Administration - MBA, Marketing and Research · (2017 - 2019)

Sir M Visvesvaraya Institute Of Technology

Bachelor of Engineering - BE, Electronics and Communications Engineering · (2015)